2020

Agras T20 Repair Training

Contents

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Product Introduction

Product Introduction - Agras T20

Product name: Agras T20

Product series: Agras Drone T Series

Product Introduction - Comparison of T series

Agras T20

Agras T16

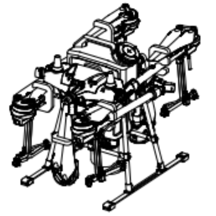
Product Introduction - Parameter Comparison

		Agras T20	Agras T16
Features	Operating Efficiency (Per Hour)	12 hectares	10 hectares
	Spray tank capacity	20L	16L
	High precision radar	Horizontal omnidirectional obstacle avoidance	Forward and backward obstacle avoidance
Flight Parameters	Total Weight (Excluding battery)	21.1 kg	18.5 kg
	Standard Takeoff Weight	42.6 kg	39.5 kg
	Max Takeoff Weight	47.5 kg (At sea level)	40.5 kg (At sea level)
	Max Thrust-Weight Ratio	1.70 (Takeoff weight of 47.5 kg)	2.05 (Takeoff weight of 39.5 kg)
	Max Power Consumption	7000 W	5600 W
	Hovering Power Consumption	6000 W (Takeoff weight of 47.5 kg)	4600 W (Takeoff weight of 39.5 kg)
	Hovering Time	15 min (Takeoff weight of 27.5 kg with a 18000 mAh battery) 10 min (Takeoff weight of 42.6 kg with a 18000 mAh battery)	18 min (Takeoff weight of 24.5 kg with a 17500 mAh battery) 10 min (Takeoff weight of 39.5 kg with a 17500 mAh battery)
Airframe Parameters	Maximum Diagonal Wheelbase	1883 mm	1895 mm
	Dimensions	2509 mm×2213 mm×732 mm (Arms and propellers unfolded) 1795 mm×1510 mm×732 mm (Arms unfolded and propellers folded) 1100 mm×570 mm×732 mm (Arms and propellers folded)	2520 mm×2212 mm×720 mm (Arms and propellers unfolded) 1800 mm×1510 mm×720 mm (Arms unfolded and propellers folded) 1100 mm×570 mm×720 mm (Arms and propellers folded)
Spraying System	Spray Tank Volume	Rated: 15.1 L, Full: 20 L	Rated: 15.1 L, Full: 16 L
	Operating Payload	Rated: 15.1 kg, Full: 20 kg	Rated: 15.1 kg, Full: 16 kg
	Nozzle Model	SX11001VS (Standard) SX110015VS (Optional, purchase separately) XR11002VS (Optional, purchase separately)	XR11001VS (Standard) XR110015VS (Optional, purchase separately)
	Max Spray Rate	SX11001VS: 3.6 L/min SX110015VS: 4.8 L/min XR11002VS: 6 L/min	XR11001VS: 3.6 L/min XR110015VS: 4.8 L/min
	Droplet Size	SX11001VS : 130 - 250 μm SX110015VS : 170 - 265 μm XR11002VS: 190 - 300 μm (Subject to operating environment and spray rate)	XR11001VS: 130 - 250 μm XR110015VS: 170 - 265 μm (Subject to operating environment and spray rate)
	Spray Width	4 - 7 m (8 nozzles, at a height of 1.5 - 3 m above crops)	4 - 6.5 m (8 nozzles, at a height of 1.5 - 3 m above crops)
	Measurement Range	0.25 - 20 L/min	0.45 - 5 L/min

Product Introduction - Parameter Comparison

		Agras T20	Agras T16
High Precision Radar Module	Model	RD2428R	RD2418R
	Operating Frequency	CE (Europe) / FCC (United States): 24.00 GHz - 24.25 GHz MIC (Japan) / KCC (Korea): 24.05 GHz - 24.25 GHz	SRRC (China) / CE (Euope) / FCC (United States) 24.00 GHz - 24.25 GHz MIC (Japan) / KCC (Korea): 24.05 GHz - 24.25 GHz
	Power Consumption	18 W	15 W
	Obstacle Avoidance System	Obstacle sensing range: 1.5 - 30 m FOV: Horizontal 360° Vertical: ±15° Working conditions: Flying higher than 1.5 m over the obstacle at a speed lower than 7 m/s Safety distance: 2.5 m (Distance between the front of propellers and the obstacle after braking) Obstacle avoidance direction: Horizontal omnidirectional obstacle avoidance	Obstacle sensing range: 1.5 - 30 m FOV: Horizontal ±50°, Vertical 0°-10° Working conditions: Flying higher than 1.5 m over the obstacle at a speed lower than 7 m/s Safety distance: 2.5 m (Distance between front of propellers and obstacle after braking) Obstacle avoidance direction: Forward or backward obstacle avoidance depending on the direction of flight
Battery	Model	AB3-18000mAh-51.8V	AB2-17500mAh-51.8V
	Capacity	18000mAh	17500mAh
Remote Controller	Model	RM500-AG	GL300N
	Operating Frequency	2.4000 GHz - 2.4835 GHz 5.725 GHz - 5.850 GHz	2.4000 GHz - 2.4835 GHz 5.725 GHz - 5.850 GHz (Japan, Russia and Ukraine do not support this frequency band)
	Effective Transmission Distance (Unobstructed, free of interference)	SRRC / MIC / KCC / CE: 3 km FCC: 5 km	SRRC / MIC / KCC / CE: 3 km NCC / FCC: 5 km
	Equivalent Isotropically Radiated Power (EIRP)	2.4 GHz SRRC / CE / MIC / KCC: 18.5 dBm; FCC : 25.5 dBm; 5.8 GHz SRRC / FCC: 25.5 dBm;	2.4 GHz: SRRC / CE / MIC / KCC: < 20 dBm FCC / NCC: < 26 dBm 5.8 GHz: SRRC / NCC / FCC: < 26 dBm KCC / CE: < 14 dBm
	Power Consumption	18 W	16 W

Product Introduction – In the Box



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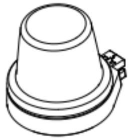
Aircraft (With Spraying System and Propeller Holders)

× 1

Remote Controller

× 1

Remote Controller Lanyard



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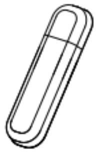
RTK Dongle



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4G Dongle

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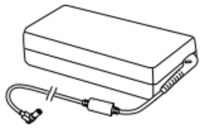
Propeller Holder*



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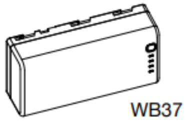
Power Adapter
電源轉接器



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Remote Controller Intelligent Battery

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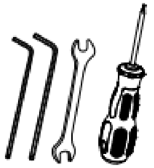
Intelligent Battery Charging Hub



Tools

3.0 mm Hex Key
4.0 mm Hex Key
Double-Ended Wrench
Double-Headed Hex Screwdriver

Screws, Nuts, and Other Accessories*

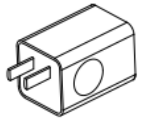
Frame Arm Pin A
Frame Arm Pin B
Hose Nuts A
Hose Nuts B
Propeller Washers
Spray Tank One-Way Valve Sealing Ring
M3×8 Screws, M3×10 Screws, M4×15 Screws, M5×10 Screws

× 1

Charging Hub AC Power Cable

× 1

USB-C Cable



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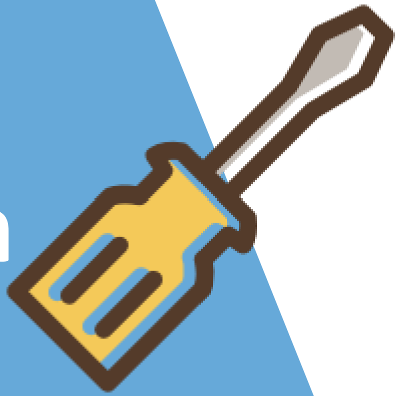
USB Charger

Manuals

In the Box
Disclaimer and Safety Guidelines
Quick Start Guide

02

Material Identification

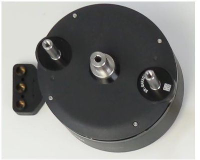


Material Identification - Module Identification

1

2

3



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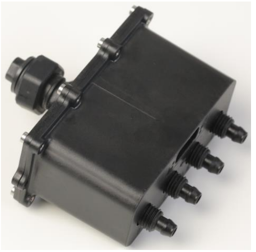

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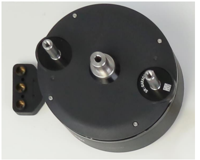
Material Identification - Module Identification

Aerial-electronics module

Spraying module

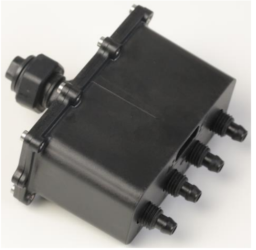
Motor



ESC

Radar

Four-channel flow meter



Delivery pump

Spray lance

Electromagnetic exhaust valve

RTK antenna module

Half-thread locking sleeve

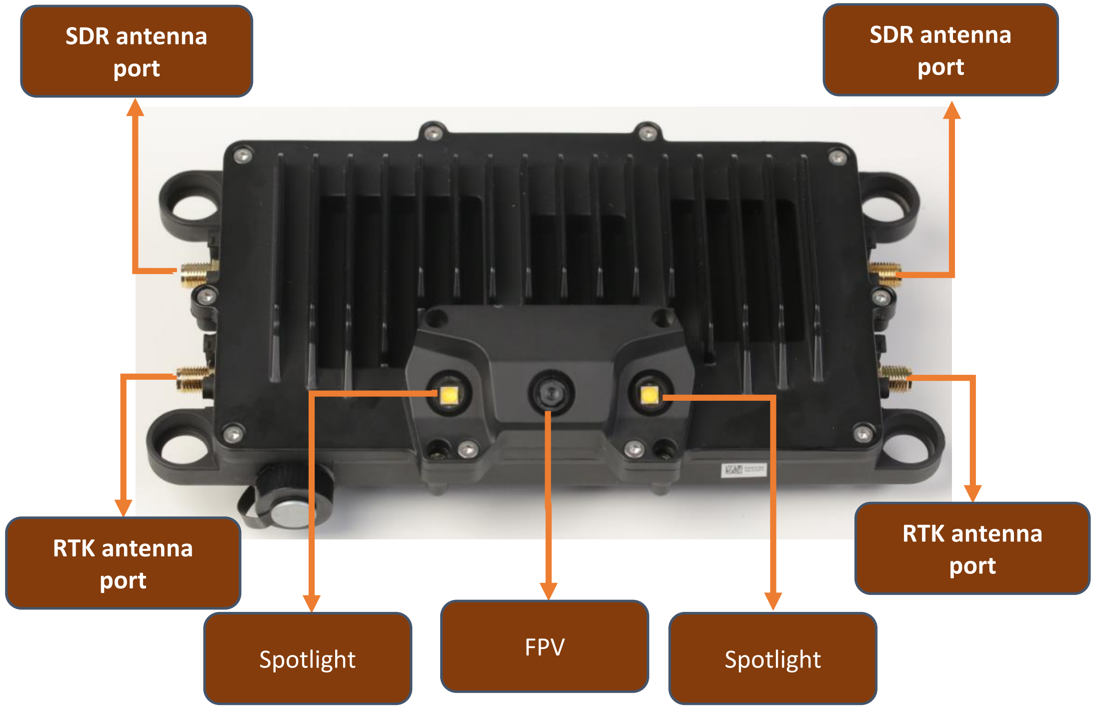
Front frame_aircraft arm cover

Front frame upper cover

Aricraft arm

Front frame

Material Identification - Front Side of Aerial-electronics Module



Agras T20 aerial-electronics module

Agras T16 aerial-electronics module

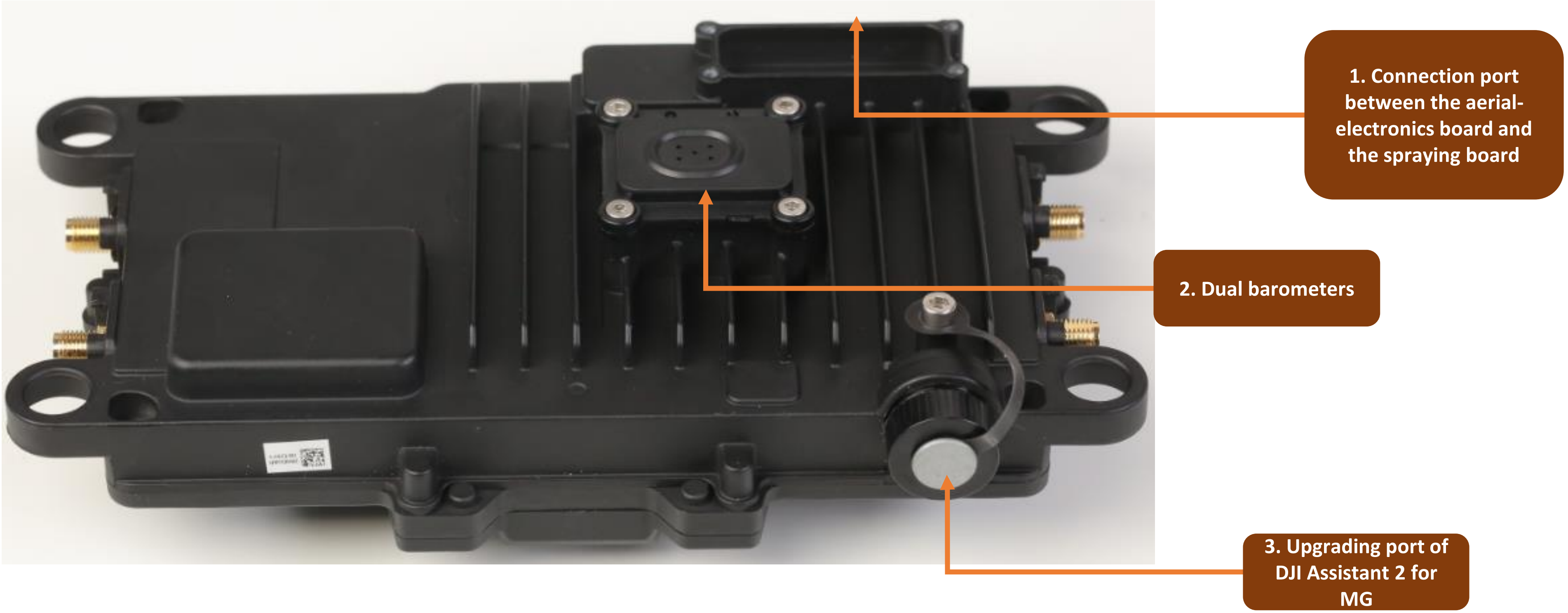
(1) The **aerial-electronics module** is the core module of the flight controller for flight control, which integrates flight algorithm, image transmission and flight control into a module. It controls various parts of the aircraft (Flight controller module, SDR module, RTK/GPS module, data protector module, dual IMUs, dual barometers, FPV)

Differences:

Label head of Agras T16 aerial-electronics board: 0UR

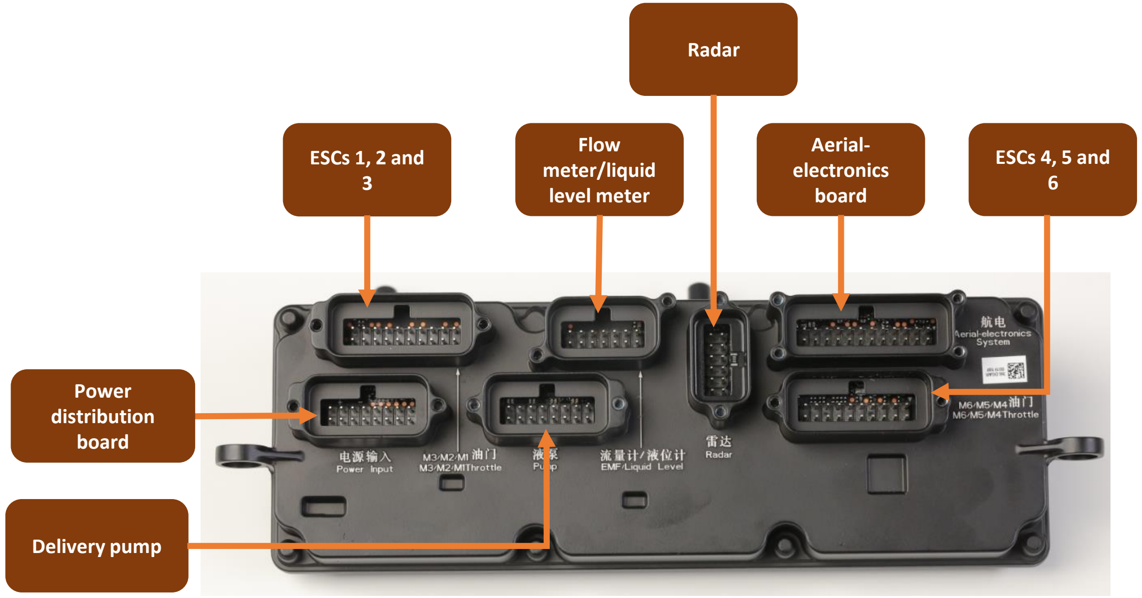
Label head of Agras T20 aerial-electronics board: 39H

Material Identification - Back Side of Aerial-electronics Module

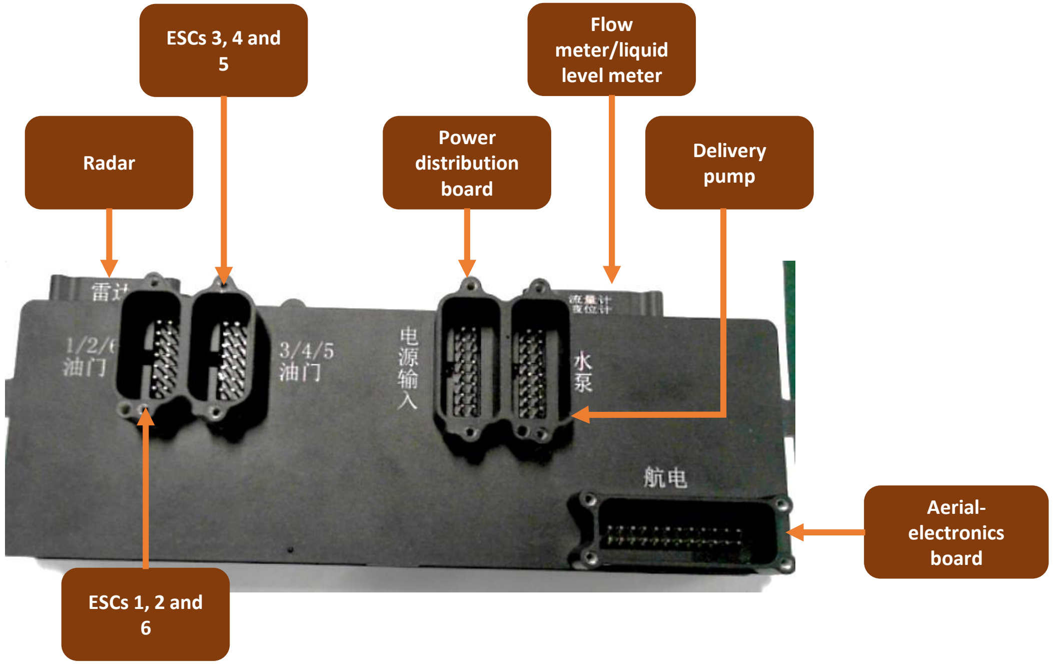


The position marked by figure 1 is the connection port between the aerial-electronics board and the spraying board.
The position marked by figure 2 is the barometer module. It is used to obtain the current pressure signal and send the pressure value to the aerial-electronics board.
The position marked by figure 3 is the upgrading port for DJI Assistant 2 for MG which is used for update the aircraft firmware.

Material Identification—Spraying Board Module



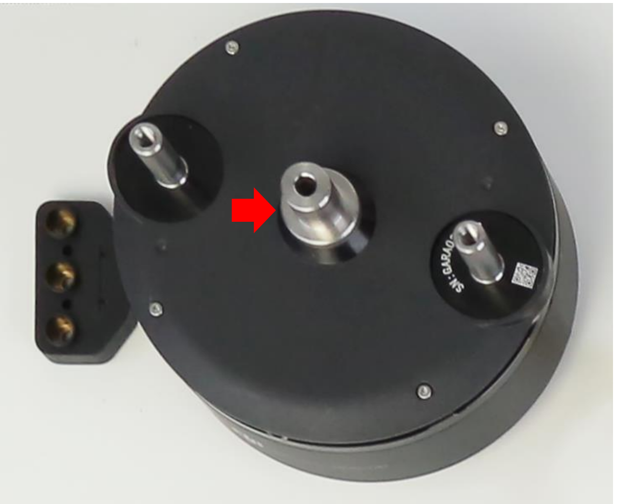
Agras T20 spraying board module



Agras T16 spraying board module

(2) The **spraying board** is the control system for the core spraying function of the aircraft. It is used to control the delivery pump and the amount of pesticides sprayed. The compass board is inside the spraying board.

Material Identification—Motor Module



Agras T 20 motor

Agras T16 motor

(3) The **motor** is the executive component for flights. By driving the propellers, the motor blows downward and provides the aircraft with rising propulsion. It also helps with spraying pesticides.

Differences:

1. The bottom bearing colors are different. The Agras T20 motor bearing is **silver** and the Agras T16 motor bearing is **black**.
2. The **middle parts** are different, as shown by the red arrow.

Material Identification—ESC Module

Agras T20 ESC

Agras T16 ESC

(4) **ESC:** The ESC board converts throttle information of the flight control to motor speed information. It controls motor rotation and provides the motor speed to the flight controller. It also provides overcurrent protection.

The ESC modules of the T series are the same.

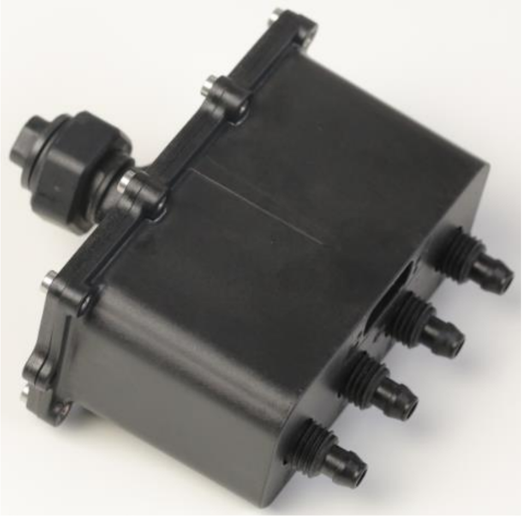
Material Identification—Radar Module

Agras T20 radar module

Agras T16 radar module

(5) **Omnidirectional rotating radar:** Agras T20 is equipped with an omnidirectional digital radar, which can detect obstacles by 360° in the horizontal direction. In addition, it allows the aircraft to bypass obstacles or avoid obstacles autonomously and support Terrain Awareness Mode to ensure safety during use. With this radar, the T20 can adapt to most environments, 24/7, thanks to its strong resistance to dust and water. The radar also acts on the spraying system.

Material Identification—Flow Meter Module



Agras T20 four-channel flow meter

Agras T16 flow meter module

(6) The **four-channel flow meter** is a high-precision key element for spraying. Each channel is responsible for calibrating and feeding the pesticide sprayed by each delivery pump for precise flow control of each delivery pump pipe. Each pump has 1/4 of the flow set by the user, ensuring an even flow sprayed by each nozzle.

Material Identification— Delivery Pump

Agras T20 delivery pump

Agras T16 delivery pump

(7) **Delivery pump:** The 6 L/min large-flow delivery pump consists of a delivery pump motor and an ESC. By receiving the signal of the spraying board and controlling the spraying effect of the aircraft, the spray rate can be adjusted by the remote controller dial. It is the executor of the aircraft spraying system, which spays pesticide through pressure control.

Differences: There are no differences in appearance; they are interchangable.

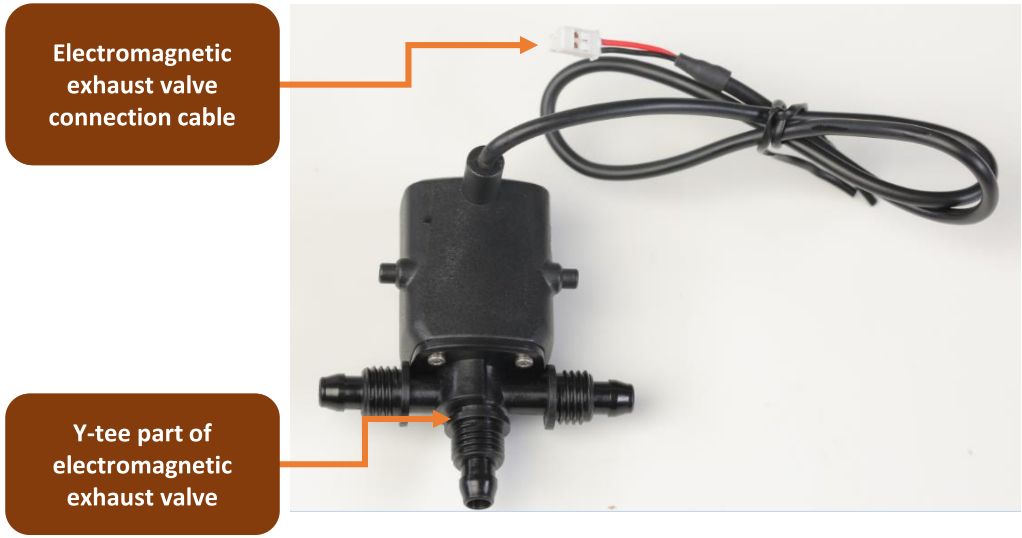
Material Identification—Spray Lance

Agras T20 spray lance part

Agras T16 spray lance part

(8) **Spray lance and electromagnetic exhaust valve**: The left spray lance extends inwards and optimizes the overall spraying effect. The electromagnetic exhaust valve has an automatic pressure relief function, which can automatically exhaust the air in the water pipe. When the delivery pump is turned on, air will be automatically exhausted. When the delivery pump is disabled, the electromagnetic exhaust valve will stop working.

Material Identification—Electromagnetic Exhaust Valve



Electromagnetic exhaust valve

Agras T16 exhaust valve

(9) **Electromagnetic exhaust valve:** The electromagnetic exhaust valve has automatic pressure relief function, which can automatically exhaust the air in the water pipe. When the delivery pump is enabled, automatically exhaust air. When the delivery pump is disabled, the electromagnetic exhaust valve stops working.

If the electromagnetic exhaust valve cannot be enabled normally, remove the screws outside the electromagnetic exhaust valve to open it.

Material Identification—RTK Antenna Module

Agras T20 RTK antenna module (Without feeder)

Agras T16 RTK antenna module

(10) **RTK antenna:** RTK module provides the flight control system with reliable information such as the aircraft position, direction and speed to ensure flight stability and accurate operation. The RTK antenna close to aircraft arm 6 is the No. 1 master antenna, and the RTK antenna close to aircraft arm 2 is the No. 2 auxiliary antenna.
Differences: Different appearance

Material Identification—Remote Controller

T20 Smart controller (RC) 2.0

Agras T16 remote controller

Brand-new remote controller with innovative productivity
The brand-new remote controller has a greatly improved performance when compared to the previous generation. The DJI MG app has been upgraded completely for smoother running of the system and simpler interface operation. Standard RTK high-precision positioning module achieves centimeter-level operation planning. A 5.5-inch CrystalSky is provided, allowing for clear display and simple operation even under strong sunlight. The remote controller supports external batteries. The overall battery life has doubled compared with the last generation, allowing it to meet the requirements for long-term and high-intensity operation.

Agras T20 remote controller: DJI Smart Controller upgrading + Expansion Module
Agras T16 remote controller (RC) : Conventional MG series remote controller

03

Assembly and Disassembly & Notices

Assembly and Disassembly & Notices

No.	Tools
1	3.0 mm hexagonal socket screwdriver
2	2.5 mm hexagonal socket screwdriver
3	2.0 mm hexagonal socket screwdriver
4	1.5 mm hexagonal socket screwdriver
5	Phillips screwdriver
6	Tweezers (metal)
7	Nipper pliers
8	Anti-static tweezers
9	Crowbar
10	Cutting pliers
11	10 mm/7 mm sleeves
12	T6 Torx screwdriver

Assembly and Disassembly & Notices

(Check the repair video)

Product	Link	Password
T30&T10	https://pan-sec.djicorp.com/s/baYy6WzKakbDMLZ	gkas123
T20	https://pan-sec.djicorp.com/s/FGH9rHAYMdJjB9F	gkas123
T16	https://pan-sec.djicorp.com/s/KYAjLiEPAnjnfxF	gkas123
Spreader （MG&T)	https://pan-sec.djicorp.com/s/AGtyx8Pq7tDYRPt	gkas123
MG series	https://pan-sec.djicorp.com/s/Q24e9sTGtzCwHQZ	gkas123

Assembly and Disassembly & Notices

The RTK antenna module is fixed by the three screws, one of which has a long screw bit.

The sleeve needs to be removed or installed for assembly and disassembly of fixing piece of the rear frame: 7 mm sleeve

Assembly and Disassembly & Notices

Please distinguish the L-shape aluminum tube.

1. Cable clamp
2. Nameplate positioning hole

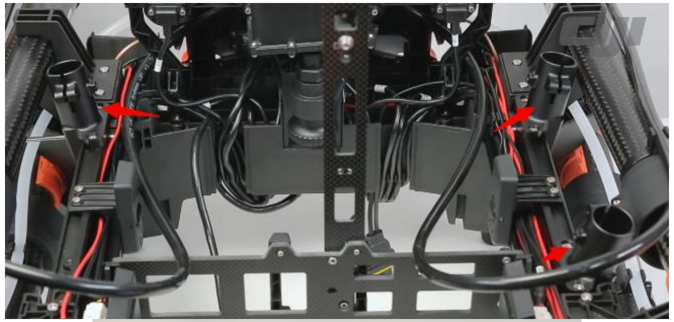
Disassembly of front and rear frame aircraft arm connectors:
A rubber hammer is required due to a firm connection.

Assembly and Disassembly & Notices

Attach insulated rubber tape on the connection cable of the electromagnetic relief valve.

There are no restrictions on the assembly and disassembly sequence of the half-thread locking sleeve.

Assembly and Disassembly & Notices



The landing gear supporting rods face outwards.
The front right/rear left supporting rods are interchangeable.
The front left/rear right supporting rods are interchangeable.

Please distinguish the front and rear supports of the radar.
The front support has a cable knob and arrow.

Assembly and Disassembly & Notices

Ensure that the delivery pump is facing outwards and avoid misassembly.

Assembly and Disassembly & Notices - Summary

Optimization of front, middle and rear frame structures:
1. The Agras T20 aircraft frames (such as front frame, rear frame and middle frame) are smaller than those of the Agras T16; assembly and disassembly methods are also simpler.
2. The assembly and disassembly of front and rear frames have been improved and the aircraft connectors have been reduced. There are two sliders on the left and right of the middle frame for the T16, and the sliders and middle frame in the T20 have been integrated for convenient operation.
3. The upper and lower covers of the front and rear frames are locked by screws instead of adhesive, making operation more convenient.

Optimization of aircraft arm structures:
1. A special mark has been attached on the connection part of the frame arm carbon tube and the aircraft to remind users to lock the sleeve.
2. The sleeve is designed with left and right locking structures. There are no restrictions on the assembly and disassembly sequence of the half-thread locking sleeve.
3. For better coverage of the spraying area and accurate flow control, the spray lance is designed with extended horizontal length and a electromagnetic flow meter has been added.

Optimization of landing gear and radar structures
1. The landing gear is made more wear resistant to reduce loss from crashes.
2. The fixing part between the landing gear and the aircraft is fixed with stronger screws.
3. The radar is locked with bilateral symmetry connectors, which simplifies repair and reinforces radar installation.

Optimization of electronic modules and screws:
1. The connection cables of all modules have uniform screw models.
2. The spraying module is vertically mounted for easy repair e and operation.
3. The sealing ring at the area of connection between the cable and the module is optimized for convenient repair and operation.

Assembly and Disassembly & Notices - Remote Controller

Lower Shell Disassembly

There are buckles between the upper and lower shells which are extremely firm after assembly. Remove all fixing screws during disassembly. Clasp the lower shell with your hand and remove it.

Front Decorative Cover Disassembly

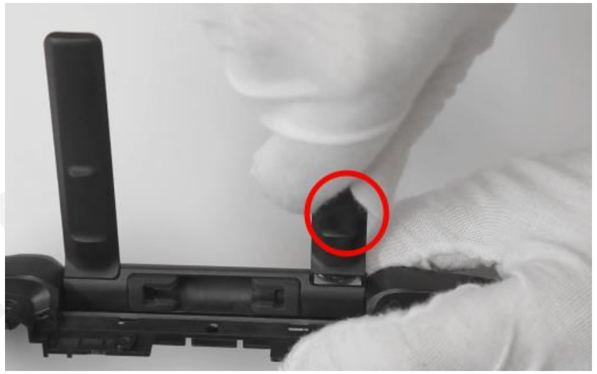
Before removing the materials from the front shell, the front decorative cover needs to be removed. Remove the two screws in the decorative cover before removing the front cover.

Front Decorative Cover Disassembly: Gently pry off the material with a crowbar and remove it forcibly.

Assembly and Disassembly & Notices - Remote Controller

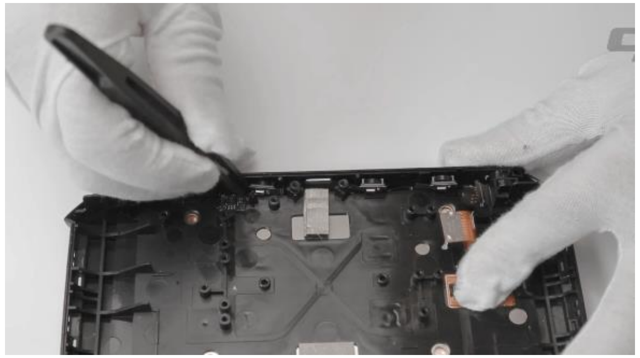
Antenna Sleeve Disassembly

The antenna sleeve disassembly involves the replacement of the image transmission antenna. During assembly, it is attached firmly with glue. Heat the antenna sleeve with a heat gun (within minimum temperature and air volume). Wait and observe until it is slightly misshapen. Shake the antenna sleeve and remove it. The removed antenna sleeve will be scrapped and not used again. Make sure that not to heat the antenna's base structures.



MIC Board Disassembly

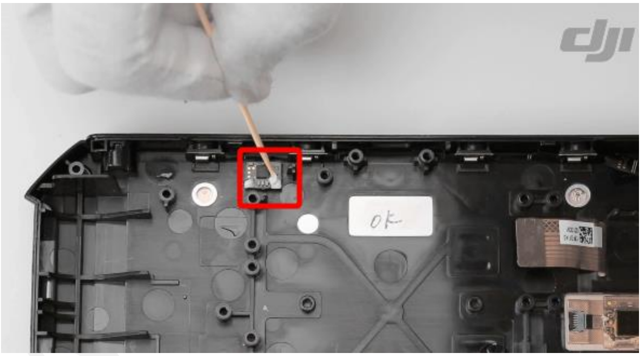
The MIC board has four fixing pieces, which are small buckles. Observe the buckle position and eject it using tweezers.



Remote Controller Screen Protection

This does not involve screen replacement. During turnover, damage assessment and maintenance, be careful not to scratch the screen. After materials are received, a protection film (material number: YC.BZ.JJ000589.01) can be attached for protection.

Assembly and Disassembly & Notices - Remote Controller



MIC Board Assembly

When the MIC board is assembled before it is shipped out from the factory, it has gone through a hot melting process. Please fix it with white glue before repair and maintenance.

Front Decorative Cover and Light Guide Beam

Before scrapping the front decorative cover, please remove the light guide beam from the front decorative cover for reuse.

Left and Right Rubber Pads

1. Stick the left and right rubber pads. Please apply Kafuter glue before sticking the left and right rubber pads.
2. 5D Button Assembly. Please use Kafuter glue before replacing the 5D Button. Put a small amount of hot melt adhesive on the button of the button combination board.

Assembly and Disassembly & Notices - Remote Controller

Description of Recommended Consumables

Left and Right Rubber Pads

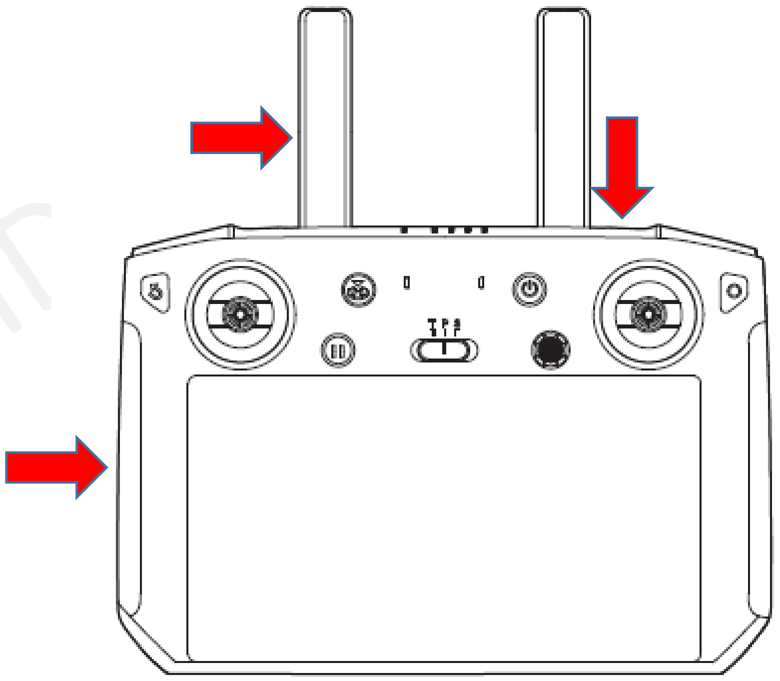
Before disassembly of the remote controller, it is recommended to replace the left and right rubber pads.

Antenna sleeve

Remove the sleeve during each replacement of the antenna. It is recommended to replace the old sleeve with a new one.

Front Decorative Cover

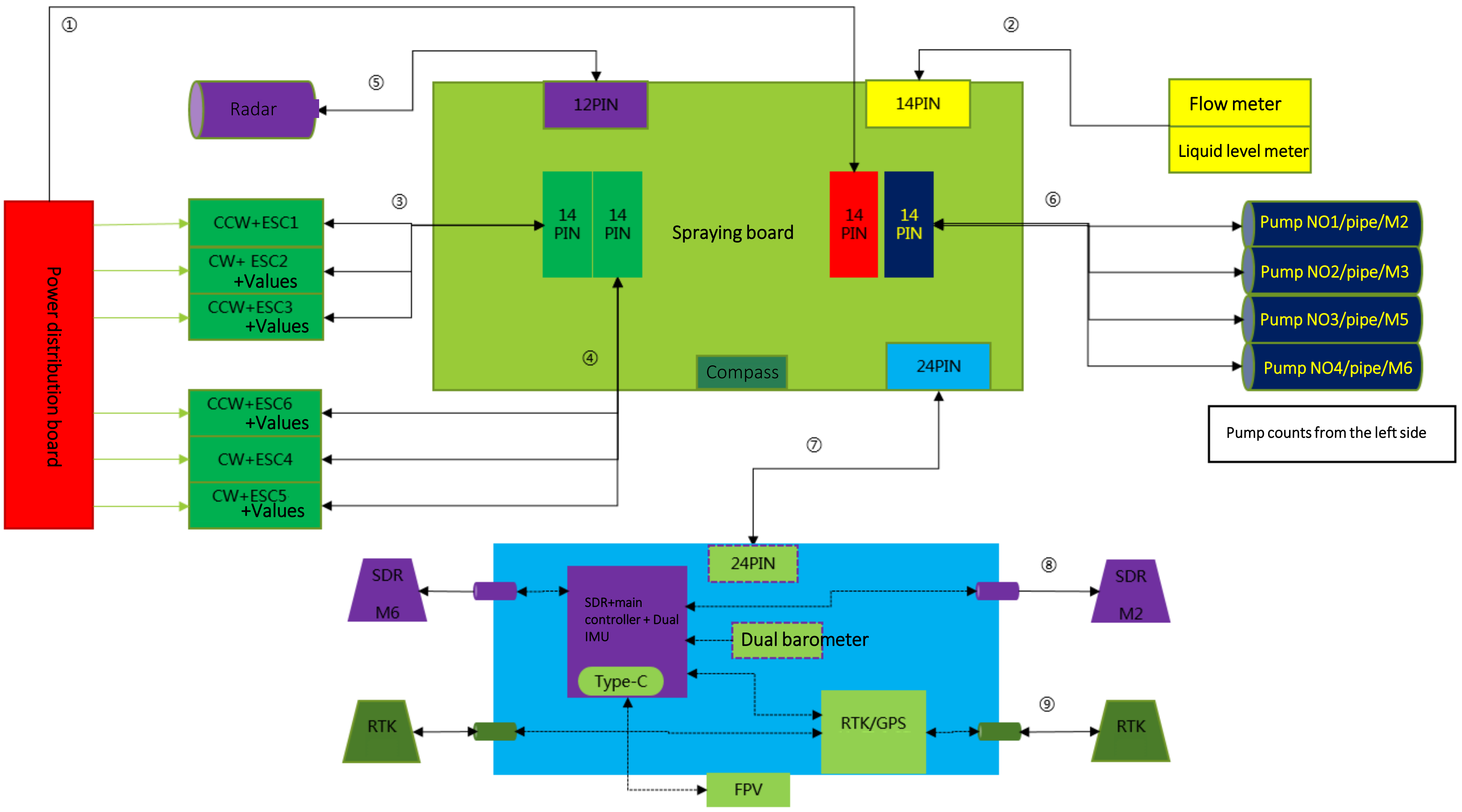
The front decorative cover should be removed forcibly. For any materials involving front cover assembly and disassembly, this component must be replaced.



04

Principle Diagram & Boards Introduction

Principle Diagram



Boards Introduction

Six Systems	17 Modules	Introduction
1, Aerial-Electronics System	Aerial-Electronics Board	The aerial-electronics board is the core module of the entire flight controller and is integrated with flight algorithms, image transmission, and flight control into one module. It is used to dispatch all resources of the aircraft.
	FPV Board	The FPV board is a transit module between the camera and the aerial-electronics board, which sends the camera information to the aerial-electronics board and drives the camera. It also provides night lighting.
	Barometer Board	The barometer module is used to obtain the current pressure signals, transmit the pressure value to the aerial-electronics board, and provide control information.
2, Image Transmission System	FPV	FPV, called first-person vision, is a camera module, which converts optical signals into electrical signals and transmits them to the remote controller via image transmission.
3, Radar System	Rotating Radar	The rotating radar is the obstacle avoidance module, and it serves as the sensor when the aircraft is flying. It detects the front, lower, and rear obstacles to prevent the drone from hurting people and crashing.

Boards Introduction

Six Systems	17 Modules	Introduction
4, Spraying System	Spraying Board	The spraying board is the control system of the aircraft's core spraying function. It controls the delivery pump and spraying volume.
	Delivery Pump ESC Board	The delivery pump ESC board is the power system of the delivery pump, and the signal of the spraying board is used to control the rotation of the delivery pump motor.
	Liquid Level Meter	The liquid level meter can obtain the amount of the pesticides in the spray tank, transfer the information to the spraying board to remind the user whether it is necessary to fill pesticides.
	Flowmeter	The flow meter is essential for high-precision spraying, which is responsible for controlling the delivery pump and the amount of sprayed pesticides. Therefore, the pump's spraying accuracy can be controlled.
	Delivery Pump	The delivery pump is the executor of the aircraft spraying system, which sprays out pesticides through rotation.
5, Positioning System	RTK/GPS Board	The RTK module provides reliable information such as the position, direction, and speed of the aircraft to the flight controller system to ensure aircraft navigation stability and operation accuracy. The RTK module contains functional circuits such as RTK, GPS, breakpoint continuation and related power management. The RTK module can track GPS L1/L2, GLONASS F1/F2, BDS B1/B2, Galileo E1/E5b frequency bands at the same time for positioning and orientation. Both GPS and RTK module uses the main antenna of RTK and receive GPS L1 and GLONASS F1 frequency band signals.
	Active Antenna Board	The active antenna board can amplify and adjust the weak GPS signal so that the RTK/GPS board can get better satellite signals.

Boards Introduction

Six Systems	17 Modules	Introduction
6, Propulsion System	Power Distribution Board	The power distribution board is a transit module, which distributes the power source to the six motors.
	ESC Board	The ESC board is responsible for converting the throttle information of the flight control into the speed information of the motor, controlling the rotation of the motor, feeding back the motor speed to the flight controller, and providing functions such as overcurrent protection.
	LED Board	The LED board is responsible for indicating the current working status of the aircraft, and informs the user of the current aircraft control information and working status through the color of the LEDs.
	Battery	The battery, an energy storage equipment, is the energy source for flight and spraying.
	Motor	The motor is an executive component of the aircraft for the flight. Drive the propellers and blow down the airflow to provide the aircraft with upward power. At the same time, it assists in spraying pesticides, making pesticides more permeable.

Introduction of Main Cables

Material Number	Material Name
YC.XC.XX000519.03	Aerial-electronics module power cable
YC.XC.XX000511.03	Power cable connecting the spraying board to the delivery pump
YC.XC.XX000513.03	Signal cable A from the spraying board to the propulsion system (aircraft arms 1, 2, and 3)
YC.XC.XX000512.03	Signal cable B from the spraying board to the propulsion system (aircraft arms 4, 5, and 6)
YC.XC.XX000510.03	Radar power cable
YC.XC.XX000518.03	Liquid level meter signal cable
YC.XC.TZ000067.04	RTK antenna feeder
YC.XC.DD000348.03	Power distribution board power cable (negative) (M4/5/6)
YC.XC.DD000349.03	Power distribution board power cable (positive) (M1/2/3)
YC.XC.DD000347.03	Power distribution board power cable (positive) (M4/5/6)
YC.XC.DD000350.03	Power distribution board power cable (negative) (M1/2/3)
YC.XC.XX000515.03	Signal cable from the power distribution board to the spraying board power supply

05

Malfunction Assessment and Analysis

Malfunction Assessment and Analysis--- Aerial-Electronics Module

Component	Failure	Reason Analysis	Solution
Aerial-electronic core board (combined with the flight controller and the SDR image transmission).	The DJI Assistant 2 for MG cannot be connected.	1. The firmware is lost.	Re-burn the firmware.
		2. The DJI Assistant 2 for MG version does not match.	Download the DJI Assistant 2 for MG of the corresponding version on the DJI official website.
		3. The computer drive installation fails.	Reinstall the drive or DJI Assistant 2 for MG or use a different computer.
		4. Poor connection	Replace the Type-C cable or the USB-C port.
	Update failure	1. The firmware is lost.	Check the small component firmware version, burn the firmware and update again.
		3. There is an error with the network.	Ensure that the network is without issue before updating.
		4. The aerial-electronics system is not completely connected.	Ensure that all modules are completely connected.
		5. Check the corresponding updating error code.	Check the corresponding component of the code.
	The aircraft hovers and drifts.	1. There is an error with the IMU calibration.	Perform the IMU (6-directional) calibration.
		2. When the satellite search is normal and the motor works normally, check whether the aerial-electronics system is abnormal (There is no obstruction or abnormal sounds. The rpm value is normal).	Replace the aerial-electronics module.
	The aircraft deviates from the straight line.	1. There is an error with the compass calibration.	Re-calibrate the compass or replace the spraying module.
		2. When the compass calibration is normal, check whether the aerial-electronics module is abnormal.	Replace the aerial-electronics module.
	There is an error with the master remote controller data (microSD card).	Power off the aircraft for 5 min and then power it on to see if it is has returned to normal.	Replace the aerial-electronics module.
	There is an error with the master remote controller data (IMU).	Perform the IMU (6-directional) calibration.	Replace the aerial-electronics module.
	There is water damage.	The liquid contact indicator has turned red or the boards are corroded.	If the aerial-electronics module is malfunctioning or the board pieces are corroded, replace the aerial-electronics module.
	The aircraft cannot be linked.	1. Check whether the remote controller matches with the aircraft firmware.	Update the remote controller and the aircraft to the same firmware version or update them both to the latest version.
		2. Perform an aerial-electronics module test with the virtual machine.	Replace the aerial-electronics module.
		3. Check whether the power distribution board port is abnormal.	Replace the power distribution module.
	The transmission distance is short or there is no image transmission. Or, the screen is frozen.	1. Check whether the remote controller matches with the aircraft firmware.	Update the remote controller and the aircraft to the same firmware version or update them both to the latest version.
		2. Check whether the aerial-electronics module, SDR antenna and remote controller antenna and feeder are damaged. Perform a coupling testing for the entire device. Check whether the antenna and links are without issue.	Replace the SDR antenna
		3. Perform a virtual machine test for the aerial-electronics module.	If the aerial-electronics board is malfunctioning, replace it.
		5. Replace the remote controller or the SDR board for cross testing and check which component is damaged.	Replace the remote controller or the SDR board.
	There is water damage.	The liquid contact indicator has turned red.	Replace the aerial-electronics module.
	Country code display (N/A)	The country code is lost.	Return it to the factory and refresh the country code.
Barometer	The aircraft height determination is incorrect or abnormal.	Check and test whether the aerial-electronics module barometer is normal with a virtual machine.	Replace the aerial-electronics module.

Malfunction Assessment and Analysis--- Aerial-Electronics Module

Component	Failure	Reason Analysis	Solution
FPV module board and camera lens	There is no image transmission.	1. Check whether the remote controller matches with the aircraft firmware.	Update the remote controller and the aircraft to the same firmware version or update them both to the latest version.
		2. Perform an aerial-electronics module test with a virtual machine.	Replace the aerial-electronics module.
		3. Perform a coupling test for the entire machine.	Replace the aerial-electronics board and perform damage assessment on the core board.
		4. Check whether the connection cable that connects the FPV board and the aerial-electronics board is damaged.	Replace the connection cable that connects the FPV board and the aerial-electronics board.
		5. Replace the camera lens for cross testing and check whether the camera lens is damaged.	Replace the aerial-electronics module and perform damage assessment on the camera lens.
		6. Replace the FPV board for cross testing and check whether it is damaged.	Replace the aerial-electronics module and perform damage assessment on the FPV board.
	The image transmission screen freezes.	1. Perform a virtual machine test for aerial-electronics module.	Replace the aerial-electronics module.
	The spotlight does not light up.	1. Perform a virtual machine test for aerial-electronics module.	Replace the aerial-electronics module.
		2. The spotlight is not turned on in the app or the brightness is abnormal.	Turn on the spotlight in the app or replace the spotlight.
		3. Check whether the connection cable that connects the FPV board and the aerial-electronics board is damaged.	Replace the connection cable that connects the FPV board and the aerial-electronics board.
RTK/GPS/Data protector module	The GNSS satellite research fails or the satellite is lost.	1. There is environmental disturbance or the firmware is inconsistent.	Try in a different environment or update the firmware.
		2. The RTK antenna contact is poor.	Reinsert the RTK antenna.
		3. Replace the RTK antenna to cross test and check whether it is damaged.	Replace the RTK antenna.
		4. Perform a satellite test for the aerial-electronics module with a virtual machine.	If the aerial-electronics module is malfunctioning, replace it.
		5. Replace the RTK board to cross test and check whether it is damaged.	Replace the RTK board.
	The aircraft will drift or frequently switch the attitude in GPS mode.	1. There is environmental disturbance or the firmware is inconsistent.	Try in a different environment or update the firmware.
		2. The RTK antenna contact is poor.	Reinsert the RTK antenna.
		3. Replace the RTK antenna to cross test and check whether it is damaged.	Replace the RTK antenna.
		4. Perform a satellite test for the aerial-electronics module with a virtual machine.	If the aerial-electronics module is malfunctioning, replace it.
		5. Replace the RTK board to cross test and check whether it is damaged.	Replace the RTK board.
		6. Replace the spraying board to cross test (Test the spraying board compass).	Replace the spraying board.
	There is an error report for the RTK angle.	Ensure that the left and right antennas are connected correctly.	Check the antenna connection sequence.
	Switch the RTK to the GPS mode.	1. Check whether the remote controller and the aircraft firmware match.	Update the remote controller and aircraft firmware.
		2. Check whether the base station is too far from the aircraft.	Move the aircraft closer to the base station.
		3. Test the aerial-electronics board with a virtual machine.	If the aerial-electronics board is malfunctioning, replace it.
	The number of RTK satellites is not stable.	1. There is environmental disturbance or the firmware is inconsistent.	Replace the satellite environment or update the firmware to the latest version.
		2. Perform a satellite test for the aerial-electronics module.	If the aerial-electronics module is malfunctioning, replace it.
	There is no power supply capacity or the flight time is too short.	1. Check whether the connection cable that connects the RTK board and the master remote controller board is damaged.	Replace the cable.
		2. Check whether the RTK board capacitance has poor contact.	Replace the RTK board.

Malfunction Assessment and Analysis- Spraying Module + Delivery Pump + Liquid Level Meter

Component	Failure	Reason Analysis	Solution
Spraying module	The battery communication cannot be read.	1. The battery signal cable of the power distribution board is damaged.	Replace the battery signal cable.
		2. The cable that connects the spraying module and the aerial-electronics module is damaged.	Replace the cable that connects the spraying module and the aerial-electronics module.
		3. Replace it with a normal spraying module for testing.	Replace the spraying module
	The liquid level meter, flow meter, or radar is malfunctioning.	Replace the liquid level meter, flow meter, or radar for testing.	If the result is OK, replace the liquid level meter, flow meter, or radar.
		Replace the spraying module for testing.	If the result is OK, replace the spraying module.
	Compass calibration fails.	1. There is environmental disturbance.	Calibrate in a different environment.
		2. The compass module is damaged. (Test the spraying board compass).	Replace the spraying module
	There is no compass value.	The compass module is damaged. (Test the spraying board compass).	Replace the spraying module
	There is water damage.	The liquid contact indicator has turned red or the board pieces are corroded.	Replace the spraying board.
Delivery pump	The delivery pump is not connected.	1. Check whether the cable that connects the delivery pump and the spraying module is properly connected.	Replace the cable that connects the delivery pump and the spraying module.
		2. Replace the delivery pump for cross testing.	Replace the delivery pump.
		3. Check whether the spraying board is damaged.	Replace the spraying module
	During self-check and rotation, an error is reported.	1. Replace the delivery pump tube and check whether the tube or the nozzle is damaged.	Ensure that the water pipe is properly connected and the nozzle does not leak.
		2. Replace the delivery pump for cross testing.	Replace the delivery pump.
	There is water leakage.	The membrane or the structure is damaged.	Replace the delivery pump.
Liquid level meter	The spray tank pesticide reading is incorrect.	1. The liquid level meter is not mounted properly or is obstructed and does not change based on the liquid level.	Re-install the liquid level meter
		2. The magnet in the liquid level meter float is backwards.	Re-assemble the liquid level meter.
	The spray tank pesticide cannot be read.	Check whether the liquid level meter is damaged.	Replace the liquid level meter.
	The liquid level meter cable is damaged.	Check whether the cable that connects the liquid level meter and spraying module is damaged.	Replace the liquid level meter cable.

Malfunction Assessment and Analysis - Radar + Flow Meter + Electromagnetic Relief Valve

Component	Failure	Reason Analysis	Solution
Radar	The radar cannot rotate.	1. Check whether the radar cable is damaged.	Replace the radar cable.
		2. The radar cannot rotate after being powered on.	Replace the radar module.
	The rotation is obstructed.	1. Check whether there are objects inside.	Clean the interfering objects.
	The height cannot maintain.	1. Check whether the radar switch has enabled in th app.	Enable the radar switch.
		2. The height determination is incorrect or fails after the radar is powered on.	Replace the radar module.
	The obstacle avoidance fails.	1. Check the obstacle avoidance instruction and alarm sound in the DJI MG2 app.	Enter obstacle avoidance mode.
		2. The obstacle avoidance is incorrect or fails after the radar is powered on.	Replace the radar module.
	There is no prompt in obstacle avoidance mode.	Check whether the radar firmware is updated to the latest version.	Update the firmware.
	There is water damage.	The internal board is corroded and damaged due to oxidation.	Replace the radar module.
Flow meter	The flow meter information cannot be read.	1. Check whether the cable is damaged or has poor contact.	Reinsert or replace the cable.
		2. Check whether the spraying board is damaged.	Replace the spraying board.
	The flow information is incorrect.	1. Check whether the flow meter is damaged.	Replace the flow meter.
		2. The flow meter fails to be calibrated.	Calibrate the flow meter.
	The flow meter cannot read the DJI Assistant 2 for MG.	1. The flow meter cable is damaged.	Replace the flow meter cable.
		2. The flow meter main board is damaged.	Replace the flow meter module.
Electromagnetic relief valve	There is a leak.	1. The connection cable is damaged.	Replace the electromagnetic relief valve module.
		2. The electromagnetic relief valve is damaged.	Replace the electromagnetic relief valve.
	There is water damage.	The liquid contact indicator has turned red.	Replace the electromagnetic relief valve.

Malfunction Assessment and Analysis - Power Distribution Board + Motor + ESC + LED

Component	Failure	Reason Analysis	Solution
Power distribution board	Unable to power on	1. Check whether the power distribution board is damaged.	Replace the power distribution board.
		2. Check whether the power cable has short-circuited or disconnected.	Replace the power cable.
	The battery communication is poor.	1. Check whether the power cable is short-circuited or disconnected	Replace the power cable.
		2. Check whether the cable is damaged.	Replace the battery signal cable.
		3. Replace with another normal battery for cross testing.	Replace the battery.
		4. Replace the spraying board for cross testing.	Replace the spraying board.
		5. Replace the aerial-electronics module for cross testing.	Replace the aerial-electronics module.
Motor / ESC	All motor alarms cannot be enabled.	1. Check whether the cable is properly connected.	Replace the connection cable.
		2. Check whether there is an abnormal prompt in the app.	Check the corresponding component based on the prompt.
		3. Check whether the spraying board is damaged.	Replace the spraying module
		4. Check whether the aerial-electronics board is damaged.	Replace the aerial-electronics module.
	No. 1, 2, and 3 motors report a warning that it cannot be enabled.	1. Check whether the cable port is damaged.	Reinsert or replace the cable.
		2. Check whether the spraying board is damaged.	Replace the spraying board.
	No. 4, 5, and 6 motors report a warning that it cannot be enabled.	1. Check whether the cable port is damaged.	Reinsert or replace the cable.
		2. Check whether the spraying board is damaged.	Replace the spraying board.
	The single motor cannot be enabled.	1. Check whether the ESC firmware is normal.	Re-burn the ESC firmware.
		2. Check whether the corresponding cable is damaged.	Replace the cable.
		3. Perform a “sweep-frequency test for the ESC board”.	If the ESC board is malfunctioning, replace it.
		4. Check whether the three-phase cable of the motor is free of glue melting or breaking. Or, check whether the riveting has broken.	If the motor is malfunctioning, replace it.
		5. Check whether the axle gear screw of the motor rotor is loose or broken.	If the motor is malfunctioning, replace it.
		6. Check whether the motor yoke is misshapen.	If the motor is malfunctioning, replace it.
	The motor has an abnormal sound.	The motor axis is misshapen or damaged.	If the motor is malfunctioning, replace it.
	The motor is obstructed.	The motor is misshapen.	If the motor is malfunctioning, replace it.
LED	The ESC light does not turn on or does not have a color.	1. The ESC board is damaged.	Replace the ESC board.
		2. The LED is damaged.	Replace the LED.
		3. The LED cable is damaged.	Replace the LED cable.

Malfunction Assessment and Analysis - Remote Controller

Component	Failure	Reason Analysis	Solution
Remote Controller	Unable to charge	1. Check whether the remote controller's built-in battery is damaged.	Replace the remote controller's built-in battery.
	Unable to power on	1. Check the external battery's battery level.	Replace the expansion module's port board.
		2. Check whether the battery cable is loose.	Tighten or replace the cable.
		3. Check the built-in battery.	Replace the built-in battery.
		4. Check the main board module.	Replace the main board
	The DJI Assistant 2 for MG cannot be connected.	1. The DJI Assistant 2 for MG version is not matched.	Download the corresponding version of DJI Assistant 2 for MG on the DJI official website.
		2. The computer drive installation fails.	Reinstall the DJI Assistant 2 for MG.
		3. Replace a normal main board for testing.	Replace the main board
	The aircraft cannot be linked.	1. Check whether the remote controller firmware matches with the Air System.	Update remote controller firmware.
		2. Replace the SDR board for testing.	Replace the SDR board.
		3. Replace a normal main board for testing.	Replace the main board
	The aircraft is lost in the specified range.	1. Check whether the antenna is damaged.	Replace the antenna.
		2. Replace the SDR board for testing.	Replace the SDR board.
		3. Replace a normal main board for testing.	Replace the main board
	The screen is abnormal.	Check whether the screen or the coaxial cable is damaged.	Replace the screen.
	The remote controller satellite search is abnormal.	Replace the remote controller screen for testing.	Replace the button combination board.
	The Wi-Fi cannot be connected.	1. Check whether the Wi-Fi antenna is damaged or is not connected.	Tighten or replace the Wi-Fi antenna.
		2. Replace a normal main board for testing.	Replace the main board
	The dial or button is malfunctioning.	Check whether the adapter plate is damaged.	Replace the Hall sensor board.
Expansion Module	The battery cannot be connected.	The type-C port of the remote controller main board is damaged.	Replace the main board
		The type-C port of the expansion module is damaged.	Replace the expansion module's port board.
	The 4G dongle cannot be connected.	The type-C port of the remote controller main board is damaged.	Replace the main board
		The type-C port of the expansion module is damaged.	Replace the expansion module's port board.
RTK Dongle	The RTK satellite search is abnormal.	1. Check the RTK port.	Tighten or replace the RTK board.
		2. Check the remote controller main board port.	Replace the remote controller main board.

Malfunction Assessment and Analysis - FAQ

1. What should I do if the DJI MG 2 app prompts that "Air in Pumps 1-4/Pumps 1-4 Flow Calibration Error" or the spraying fails when the aircraft takes off but spraying from the ground is normal?

Answer: 1. Press the spray tank close to the aircraft nose and ensure that the one-way valve of the spray tank has complete contact with the filter screen.

2. Replace the filter screen at the outlet of the spray tank.

2. What should I do if the battery cannot be powered on or charged?

Answer: 1. Ensure that there are no foreign objects in the battery port and the pin and the appearance are not misshapen.

2. Contact DJI after-sales service team and send the battery back.

3. Can I extend the flight time for Agras T16 aircraft by using the Agras T20 battery? Can it carry a payload of 20 kg?

Answer: You cannot extend the flight time for Agras T16 aircraft by using the Agras T20 battery. The aircraft can carry 20 kg but the power distribution board needs to be upgraded in order to do so.

4. Can the Agras T20 aircraft carry 20 kg when an Agras T16 battery is used?

Answer: No, the recommended payload is 15 kg.

Compatibility List

Module	T16	T20	T10	T30
Aerial-Electronics System Module	x	x	√	√
Spraying Module	x	x	√	√
RF Module	x	x	√	√
RTK Module	x	x	√	√
Power Distribution Board	x	x	x	x
Cable Distribution Board	None	None	x	x
Radar	x	x	√	√
Delivery Pump	√	√	x	x
Electromagnetic Exhaust Valves	None	x	x	x
FPV Module	x	x	√	√
Liquid Level Gauge	x	x	x	x
Upward Radar	None	None	x	x
ESC	x	x	√	√
Remote Controller	x	x	√	√

Compatibility List

Material Name	Material Number	T16	T20	T10	T30
10015 Motor (Thickness)	BC.AG.AA000137.02	√	×	×	×
10015 Motor (Thin)	BC.PR.AA000090.01	×	√	×	×
10010S Motor	BC.AG.AA000336.01	×	×	√	×
10018 Motor	BC.AG.AA000335.01	×	×	×	√
Silicone Rubber Pad	YC.XJ.QT000196.04	√	√	√	√
Teflon Gasket	YC.SJ.WS002416.03	√	√	√	√
Aluminum Alloy Propeller Holder CCW	YC.JG.QX000240.01	×	√	√	√
Aluminum Alloy Propeller Holder CW	YC.JG.QX000238.03	×	√	√	√
33-inch Propeller CCW	YC.SJ.GG000060.01	√	√	√	×
33-inch Propeller CW	YC.SJ.GG000061.01	√	√	√	×
38-inch Propeller CCW	YC.JG.ZS000813.01	×	×	×	√
38-inch Propeller CW	YC.JG.ZS000812.01	×	×	×	√

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Thank you for your participation!